THE INDUCED TOPOLOGY ON A SUBCATEGORY FROM THE BIG ZARISKI TOPOLOGY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [401, 411] and proof(s) [[413, 422]]; this spec covers the complete environment instead (statement 401-411, proof span(s) [(413, 422)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original definition 020S: sources/stacks-chow/tags.txt, source lines 2645–2645. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00VU: sources/stacks-chow/tags.txt, source lines 1160–1160. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 000R: sources/stacks-chow/tags.txt, source lines 36–36. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 000X: sources/stacks-chow/tags.txt, source lines 42–42. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let Sch_{Zar} be a big Zariski site as in Definition 020S. Let $T \in \mathrm{Ob}(Sch_{Zar})$. Let $\{T_i \to T\}_{i\in I}$ be an arbitrary Zariski covering of T. There exists a covering $\{U_j \to T\}_{j\in J}$ of T in the site Sch_{Zar} which is tautologically equivalent (see Sites, Definition 00VU) to $\{T_i \to T\}_{i\in I}$.*

Proof. Since each $T_i \to T$ is an open immersion, we see by Sets, Lemma 000R that each T_i is isomorphic to an object V_i of Sch_{Zar}. The covering $\{V_i \to T\}_{i\in I}$ is tautologically equivalent to $\{T_i \to T\}_{i\in I}$ (using the identity map on I both ways). Moreover, $\{V_i \to T\}_{i\in I}$ is combinatorially equivalent to a covering $\{U_j \to T\}_{j\in J}$ of T in the site Sch_{Zar} by Sets, Lemma 000X. □

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